Solar Marine Lanterns

DM860 Series

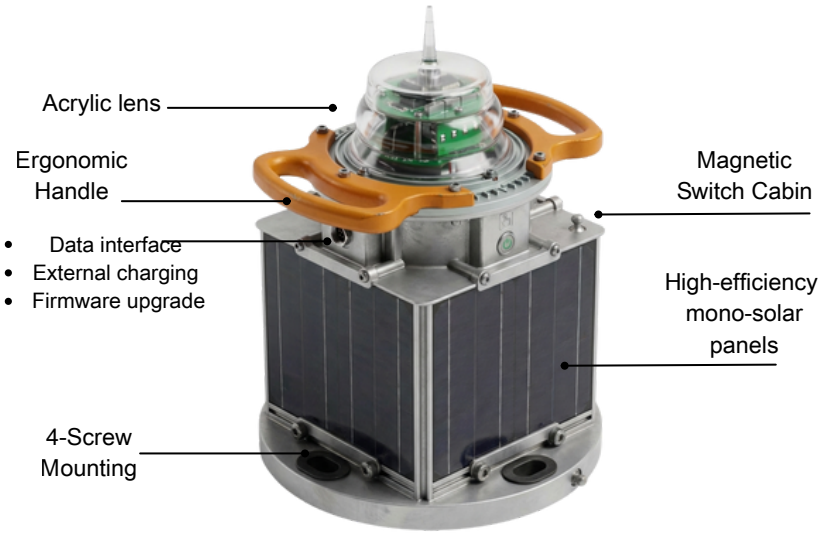
Models
- DM860
- DM860-GSM
- DM860-AIS

IALA

Standard: 256 IALA, GPS Sync, Bluetooth

Optional: Satellite Communications, AIS

The **DM860** is a compact, all-in-one 3–5 NM marine lantern designed for easy installation and reliable performance in marine navigation applications.



Key Features:

- Nominal range of 3–5 NM
- Long-life battery design supporting up to 15 years service life
- Built-in Bluetooth /GPS synchronisation
- IP68-rated enclosure for harsh marine environments
- AIS AtoN optional (Type 1 & Type 3)

Intensity Range

The DM860 provides a nominal visible range of 3–5 NM, with fine intensity adjustment to suit different navigation and marking requirements while maintaining high energy efficiency.

Standard Configuration

The DM860 is supplied as standard with RS232, RS485, Bluetooth, and GPS synchronisation, supporting 256 IALA light characters for flexible local configuration and commissioning.

AIS AtoN Capability

Optional AIS AtoN functionality supports FATDMA, RATDMA, and SOTDMA operating modes, with a communication range of 15–20 NM, among the longest in the industry.

A reinforced, segmented antenna design allows easy transport while ensuring high mechanical strength and extended range.

IP68 Submersible Design

The lantern features an IP68-rated enclosure, allowing continuous operation even under temporary submersion.

This ensures reliable performance when buoys tilt or are submerged during heavy sea conditions.

Remote Monitoring & Control (GSM)

An optional GSM telemetry module enables remote monitoring and control for near-shore applications.

The system uses a removable SIM card, allowing users to insert a local SIM or use a supplier-provided annual data plan. Custom offshore communication modules can be provided for deep-sea projects

DEXON Marine

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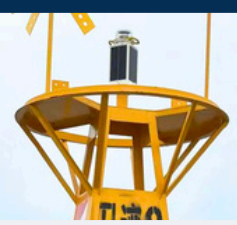

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	Specification	Details
Light Features	Light Colors	Red/Green/Yellow/White/Blue/Yellow-Blue
	Led lifespan	≥100000hours
	Vertical Divergence	Buoy Light ≥8°
	Intensity Adjustments	0–5 levels intensity settings
	Daylight Threshold	Digital photic, adjustable threshold
	Visible Range @0.74T	3-5nm
Device Features	Solar Panel	4*3.5W mono
	Lens Material	Optical Acrylic
	Base Material	Marine-grade Fiber Reinforced Plastics
	Battery Type	LiFePO4 battery
	Battery Capacity	≥288WH
	Waterproof Rating	IP68
	Parameter Configuration	RS232 / Keypad / Bluetooth
	Continuous Operation	≥25 days in continuous rainy weather
	Operating Temperature	-35°C – +65°C
AIS Features	Modulation	GMSK/FSK
	Frequency Range	156.025MHz – 162.025MHz
	AIS1 Channel	CH87B (161.975MHz)
	AIS2 Channel	CH88B (162.025MHz)
	Average Power Consumption	≤0.05W
	Transmit Power	1W 2W 5W 6W 12.5W
	Receiver Sensitivity	≤-107dBm
	Message Types	Support messages: 6, 7, 8, 12, 13, 14, 20, 21, 25
	Operating Mode	FATDMA, RATDMA, and SOTDMA modes are available, with SOTDMA as the default.
GSM Remote monitoring & control	Network	4G
	Transmit Power	2W
	GSM Remote Control & Telemetry	SIM card–based design with plug-in SIM slot
	SIM Card Options	• Supports local SIM cards for easy on-site replacement • Factory-provided international SIM card, available with annual subscription
Certification	IALA/CE/Test Reports/Product Safety Report for Sea and Air Shipment	

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